

PVD

PVD is a cancer drug treatment. It is used to treat a type of blood cancer called myeloma.

On this page

- [What is PVD?](#)
- [How PVD is given](#)
- [About side effects](#)
- [Very common side effects](#)
- [Other side effects](#)
- [Other important information](#)
- [How we can help](#)

What is PVD?

PVD is a combination of cancer drugs. It is used to treat a type of blood cancer called [myeloma](#).

PVD includes the following drugs:

- panobinostat – a [targeted therapy](#).
- [bortezomib \(sometimes called Velcade[®]\)](#) – a targeted therapy.
- dexamethasone – a [steroid drug](#).

It is best to read this information with our general information about [cancer drug treatments](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How PVD is given

During your course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before and during your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Bortezomib is an injection under the skin (subcutaneous). You usually have this in a chemotherapy day unit or clinic as an outpatient.

Your cancer team will give you the panobinostat capsules and dexamethasone tablets to take home. They may give you capsules and tablets of different strengths. Always take them exactly as they tell you to. This is important to make sure they work as well as possible for you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

Your course of treatment

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you. They may give you a copy of the plan to take home. They will explain:

- how long each cycle lasts
- when you have bortezomib injections
- when to take panobinostat and dexamethasone.

For example, a 21day (3 week) cycle of PVD may be given like this. In week 1 and 2, you have:

- bortezomib 1 or 2 times a week
- panobinostat 3 times a week
- dexamethasone 2 or 4 times a week.

Week 3 of the cycle is a rest period. During this time, you do not have any of the treatment drugs.

Your cancer team will also explain if this pattern will change. For example, usually after 8 cycles, you have fewer bortezomib and dexamethasone doses.

Having bortezomib injections

You have bortezomib as an injection into your thigh or tummy (abdomen). You have the injection in a different place each time. Tell your doctor, nurse or pharmacist if the skin in the area that was injected is:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin.

Taking panobinostat capsules

Only take panobinostat on the days you have been told to. Take the capsules once on those days. Try to take them at the same time each day. You can take panobinostat with or without food. Make sure you:

- swallow the capsules whole with a glass of water.
- do not chew, crush or open them.
- avoid eating grapefruit, star fruit or pomegranate, and avoid drinking grapefruit or pomegranate juice during your treatment. These may increase side effects.

If you forget to take panobinostat, you can still take it up to 12 hours after the time you usually take it. After this, do not take the missed dose – take your usual dose the next time it is due. Never take a double dose.

If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.

Other things to remember about your capsules:

- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- Do not throw away unused capsules. Return them to your cancer team at the hospital.

Taking dexamethasone tablets

Steroids can irritate the stomach lining. If you have indigestion or pain in your tummy, tell your doctor, nurse or pharmacist. They can prescribe drugs to help.

Take your dexamethasone with food to help protect your stomach. Some steroid tablets are coated to help reduce irritation.

Some people find it more difficult to sleep while taking steroids. Taking your dexamethasone in the morning may help.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature

- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Diarrhoea

This treatment may cause severe [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. If you have a stoma, it will be more active than usual.

Your doctor, nurse or pharmacist may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or a mild increase in stoma activity, follow their advice about:

- taking anti-diarrhoea drugs
- how much and what type of fluids to drink
- any changes to your diet that might help.

Contact the hospital straight away on the 24-hour number if any of the following things happen:

- you have diarrhoea at night
- you have uncomfortable stomach cramps
- you have diarrhoea 4 or more times in a day
- you have a moderate increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

You may need to give a stool sample, or go to hospital to have fluids through a drip or antibiotics.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Blood pressure changes – dizziness and headaches

This treatment may cause low or high blood pressure. Tell your doctor, nurse or pharmacist if you have ever had any problems with your blood pressure. If you take medicine for high blood pressure, they may need to adjust the dose.

You will have regular blood pressure checks before and during treatment. Tell your doctor, nurse or pharmacist if you:

- feel dizzy, light-headed or faint when you get up from a lying or sitting position - it may help to move slowly and drink plenty of fluids
- have any headaches.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if you:

- have sudden or severe pain in your tummy
- have signs of bleeding from the rectum (back passage)
- have black stools (poo)
- vomit up blood or have vomit that looks like coffee grounds.

Effects on the lungs

This treatment can cause inflammation of the lungs. This is called pneumonitis. Contact the hospital straight away on the 24-hour number if you notice any of these changes during treatment or after it ends:

- breathlessness
- a cough that does not go away
- wheezing
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have:

- tests to check your lungs
- treatment, if needed.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

Effects on the eyes

Your eyes may become watery and feel sore. Your doctor, nurse or pharmacist can give you advice and eye drops to help with this.

Contact the hospital on the 24-hour number if:

- your eyes get red and inflamed (conjunctivitis)
- you have pain in your eyes
- you notice any change in your vision.

Mood changes

This treatment can affect your mood. Tell your doctor, nurse or pharmacist if you have:

- feelings of anxiety or restlessness
- mood swings (moods that go up and down)
- low mood or depression.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest

- difficulty breathing.

Skin changes

This treatment can affect your skin. It might feel dry. You may develop a rash, which may be itchy. Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Effects on the thyroid

Sometimes this treatment can affect how the [thyroid](#) works. This is usually mild. You may have blood tests if needed to check how well your thyroid is working.

Effects on the brain

Rarely, this treatment causes a brain condition that can be serious. You can make a full recovery from this. But it must be diagnosed and treated quickly.

This condition can cause:

-

a headache that does not get better

- drowsiness or confusion
- changes in eyesight
- fits (seizures).

If you have any of these symptoms, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

You should not drive yourself to hospital.

Other important information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)